I- Tangents and normals

Investigation

The curve $y = x^3 - 4x$ **is shown on the graph alongside.**

Things to think about :

1. At which points is the tangent to the curve horizontal ?
2. On which intervals is the curve :
   (a) increasing
   (b) decreasing
3. At which point does the curve change its *shape* ? How can we use calculus to find this point ?

Property

Consider a curve $y = f(x)$.

If A is the point with x-coordinate a, then the gradient of the tangent to the curve at this point is

$$f'(a) = m_T.$$

The equation of the tangent is :

$$\frac{y - f(a)}{x - a} = f'(a) \quad \text{or} \quad y = f'(a)(x - a) + f(a).$$

Definition

A **normal** to a curve is a line which is perpendicular to the tangent at the point of contact.

Property

he gradient of the **normal** to the curve at $x = a$ is

$$m_N = -\frac{1}{f'(a)}.$$

Example

If $f(x) = x^2$ then $f'(x) = 2x$.

At $x = 2$,

$$m_T = f'(2) = 4 \quad \text{and} \quad m_N = -\frac{1}{f'(2)} = -\frac{1}{4}.$$

So, at $x = 2$ the tangent has gradient 4 and the normal has gradient $-\frac{1}{4}$.

Since $f(2) = 4$, the tangent has equation

$$y - 4 = 4(x - 2) \quad \text{or} \quad y = 4x - 4,$$

and the normal has equation

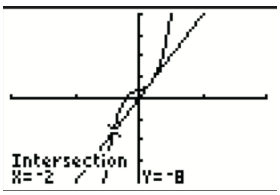
$$y - 4 = -\frac{1}{4}(x - 2) \quad \text{or} \quad y = -\frac{1}{4}x + \frac{9}{2}.$$

Example

Find the equation of the tangent to $f(x) = x^2 + 1$ at the point where $x = 1$

Example

Find the coordinates of the point(s) where the tangent to $y = x^3 + x + 2$ at $(1, 4)$ meets the curve again.



Example

Find the equations of the tangents to $y = x^2$ from the external point $(2, 3)$.

II- Increasing and decreasing functions

Property

Suppose S is an interval in the domain of $f(x)$, so $f(x)$ is defined for all x in S.

— $f(x)$ is **increasing** on $S \Leftrightarrow f(a) \leqslant f(b)$ for all $a, b \in S$ such that $a < b$.

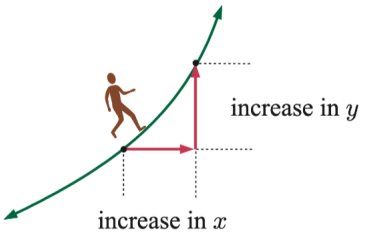
— $f(x)$ is **decreasing** on $S \Leftrightarrow f(a) \geqslant f(b)$ for all $a, b \in S$ such that $a < b$.

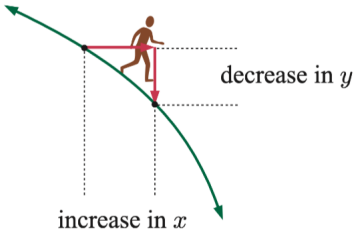
Property

Suppose S is an interval in the domain of $f(x)$, so $f(x)$ is defined for all x in S.

— $f(x)$ is **increasing** on $S \Leftrightarrow f'(x) \geqslant 0$ for all x in S

— $f(x)$ is **decreasing** on $S \Leftrightarrow f'(x) \leqslant 0$ for all x in S





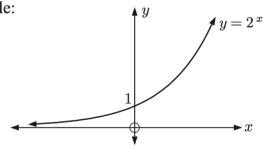
Remark. — $f(x)$ is **strictly increasing** on $S \Leftrightarrow f'(x) > 0$ for all x in S

— $f(x)$ is **strictly decreasing** on $S \Leftrightarrow f'(x) < 0$ for all x in S

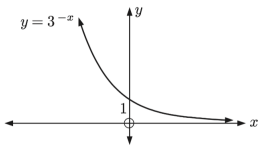
Definition

When a function is either increasing or decreasing for all $x \in \mathbb{R}$, we say it is **monotone increasing** or **monotone decreasing**.

For example:



$y = 2^x$ is strictly increasing for all x.

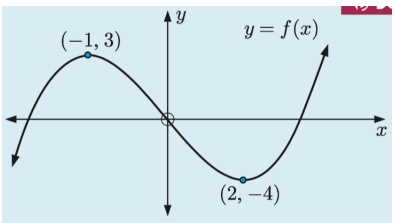


$y = 3^{-x}$ is strictly decreasing for all x.

Example

Find intervals where $f(x)$ is :

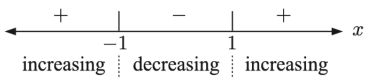
1. increasing.
2. decreasing.



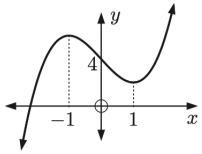
Sign diagrams for the derivative are extremely useful for determining intervals where a function is increasing or decreasing. Consider the following example :

$$\begin{aligned} f'(x) &= 3x^2 - 3 \\ &= 3(x^2 - 1) \\ &= 3(x+1)(x-1) \end{aligned}$$

which has sign diagram



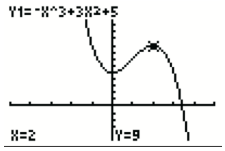
$f(x) = x^3 - 3x + 4$



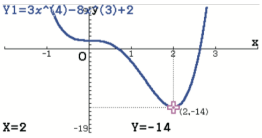
Example

Find the intervals where the following functions are increasing or decreasing :

1. $f(x) = -x^3 + 3x^2 + 5$



2. $f(x) = 3x^4 - 8x^3 + 2$



III- Stationary points

a- Extrema

Definition

A stationary point of a function is a point where $f'(x) = 0$. It could be a local maximum, local minimum, or stationary inflection.

Remark. — A global minimum is the minimum value of y on the entire domain.

— A **global maximum** is the maximum value of y on the entire domain.

Example

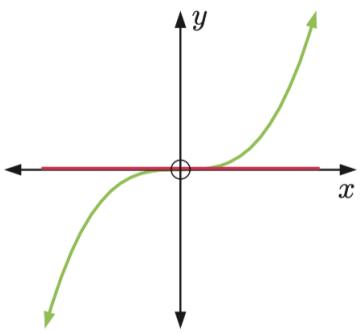
For $y = x^2$, the point $(0, 0)$ is a local minimum and is also the global minimum.

b- Stationary points of inflection

It is not always true that whenever we find a value of x where $f'(x) = 0$, we have a local maximum or minimum.

Example

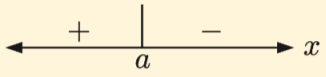
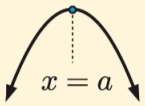
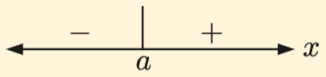
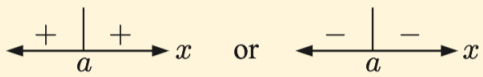
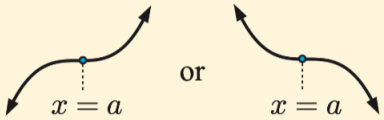
$f(x) = x^3$ has $f'(x) = 3x^2$ and $f'(x) = 0$ when $x = 0$.



Definition

It is called a **stationary inflection** (or **inflexion**) as the curve changes its curvature or shape.

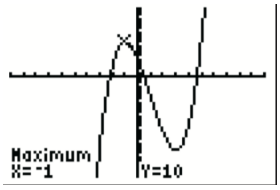
c- Sign diagrams

Stationary point where $f'(a) = 0$	*Sign diagram of* $f'(x)$ *near* $x = a$	*Shape of curve near* $x = a$
local maximum		
local minimum		
stationary inflection		

Example

Find and classify all stationary points of

$$f(x) = x^3 - 3x^2 - 9x + 5.$$



Example

Find the greatest and least value of

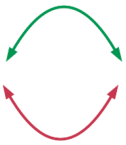
$$y = x^3 - 6x^2 + 5$$

on the interval $-2 \leqslant x \leqslant 5$.

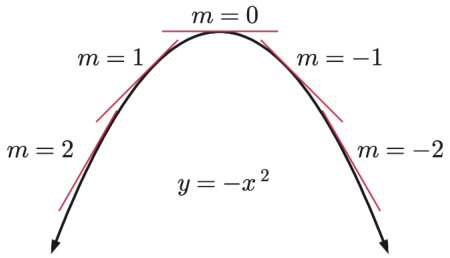
IV- Shape of a curve

Definition

When a curve, or part of a curve, has shape :

—  we say that the shape is **concave downwards**

— we say that the shape is **concave upwards**.



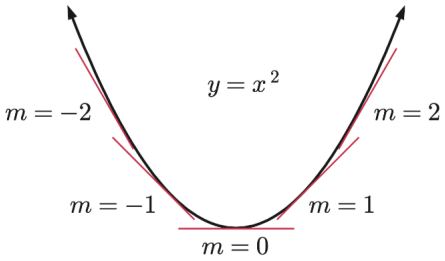
Wherever we are on the curve, as x increases, the gradient of the tangent decreases.

$\therefore$ $f'(x)$ is decreasing

$\therefore$ its derivative is negative,

which means

$$f''(x) < 0.$$



Wherever we are on the curve, as x increases, the gradient of the tangent increases.

$\therefore$ $f'(x)$ is increasing

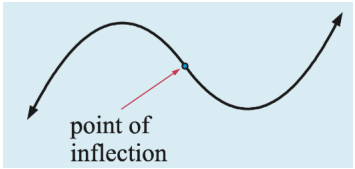
$\therefore$ its derivative is positive,

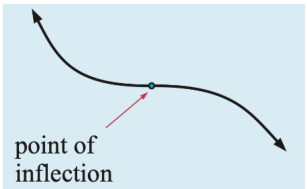
which means

$$f''(x) > 0.$$

Definition

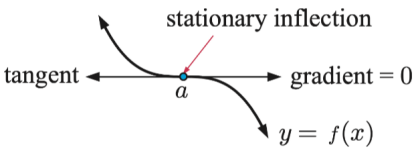
A point of inflection is a point on a curve at which there is a change of curvature or shape.





Property

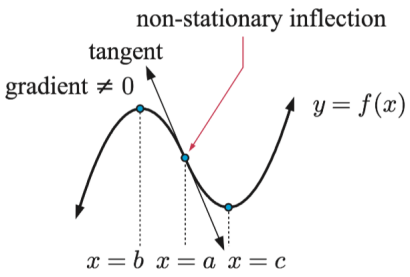
f the tangent at a point of inflection is horizontal, then this point is a horizontal or stationary inflection.



Sign diagram of $f'(x)$

Sign diagram of $f''(x)$

Remark. If the tangent at a point of inflection is not horizontal, we have a non-horizontal or non-stationary inflection.



Sign diagram of $f'(x)$

Sign diagram of $f''(x)$

Definition

The tangent at the point of inflection, also called the **inflecting tangent**, crosses the curve at that point.

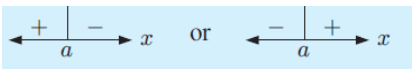
Property

There is a point of inflection at $x = a$ if $f''(a) = 0$ **and** the sign of $f''(x)$ changes at $x = a$.

The point of inflection corresponds to a change in curvature.

Property

In the vicinity of a, $f''(x)$ has sign diagram either



Example

Find and classify all points of inflection of

$$f(x) = x^4 - 4x^3 + 5.$$

Example

Consider $f(x) = 3x^4 - 16x^3 + 24x^2 - 9$.

1. Find and classify all points where $f'(x) = 0$.
2. Find and classify all points of inflection.
3. Find intervals where the function is increasing or decreasing.
4. Find intervals where the function is concave up or down.
5. Sketch the function showing the features you have found.

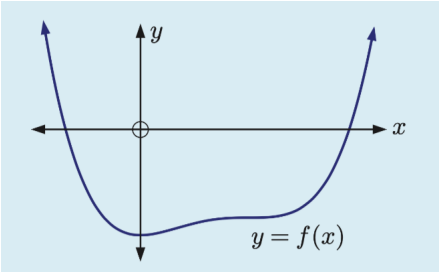
Example

Consider the function $y = 2 - e^{-x}$.

1. Find the x-intercept.
2. Find the y-intercept.
3. Show algebraically that the function is increasing for all x.
4. Show algebraically that the function is concave down for all x.
5. Use technology to help graph $y = 2 - e^{-x}$.
6. Explain why $y = 2$ is a horizontal asymptote.

Example

Using the graph of $y = f(x)$ alongside, sketch the graphs of $y = f'(x)$ and $y = f''(x)$.



Example

The graph alongside shows a gradient function $y = f'(x)$.

Sketch a graph which could be $y = f(x)$, showing clearly the x-values corresponding to all stationary points and points of inflection.

